MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FLIP-FLOP, JK, DIGITAL, KEY CODE GHK

This amendment forms a part of MIL-M-28787/296A(EC), dated 14 July 1980, and is approved for use within the Space and Naval Warfare System Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGES 4, 5, AND 6

TABLE I: Delete and substitute new table I as printed on pages 3, 4, and 5 of this amendment.

PAGE 7

* 4.4.2, delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through DD shall be performed in any order with the following exceptions. Test conditions A, B, C, D, and F are stress tests and will be performed only at 25°C ±5°C. If test conditions A, B, C, D, or F are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, F where applicable, followed by the remaining tests in the sequence."

* 4.5.1, fifth line: Delete "D" and substitute "E, G".

* 4.5.3, third line: Delete "D through" and substitute "E, G through".

* 4.5.3, sixth line: Delete "D" and substitute "E, G".

PAGE 8

6.5, second line: Delete "100°C" and substitute "85°C".

PAGE 9

6.5, Operating frequency, class I and class II: Delete "25" (two places) and substitute "18.2" (two places).

* 6.5, Output drive current, class I and class II: Delete "-400 mA" (two places) and substitute "-400 max" (two places).

PAGE 17

FIGURE 7: Delete and substitute new figure 7 as printed on page 6 of this amendment.

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FIGURE 11: Delete and substitute new figure 11 as printed on page 7 of this amendment.

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TABLE IX, Note 2, first line: Delete "voltmeter V03, point C" and substitute "item M02 (point C)".

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FIGURE 12: Delete and substitute new figure 12 as printed on page 8 of this amendment.

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* TABLE XII, Test conditions L1-L18, Initial limits, Min column: Delete "-0.1" (two places) and substitute "0.00" (two places).

PAGE 46

* TABLE XII, Test conditions L19-L24, Initial limits, Min column: Delete "-0.1" (two places) and substitute "0.00" (two places).

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TABLE XIV, Test conditions N1 through N14, Min column: Delete "10" (two places) and substitute "5" (two places).

TABLE XIV, Test conditions N15 and N16, Min column: Delete "20" (two places) and substitute "10" (two places).

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TABLE XIV, Test conditions N17 through N28, Min column: Delete "20" (two places) and substitute "10" (two places).

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TABLE XVII, Test conditions W1 through W14, Min column: Delete "75" (two places) and substitute "110" (two places).

TABLE XVII, Test conditions W1 through W14, Max column: Delete "85" (two places) and substitute "120" (two places).

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FIGURE 26, NOTE 1: Delete "PRP = 40 +2.0, -0 ns (25.0 +0, -1.2 MHz)" and substitute "PRP = 55 +4.0, -0 ns (18.18 +0, -1.23 MHz)".

PAGES 82 AND 83

TABLE XXI, Test conditions BB1 through BB24, Min column: Delete "5" (four places) and substitute "2" (four places).

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FIGURE 33, NOTE 3a: Delete "$t_{setup(1)}$ = 20 +2, -0 ns" and substitute "$t_{setup(1)}$ = 30 +3, -0 ns".

FIGURE 33, NOTE 3b: Delete "$t_{setup(0)}$ = 20 +2, -0 ns" and substitute "$t_{setup(0)}$ = 30 +3, -0 ns".

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FIGURE 35, NOTE 3b: Delete "$t_{hold(0)}$ = +2, -0 ns" and substitute "$t_{hold(0)}$ = 5 +2, -0 ns".

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TABLE I. Module electrical requirements.

	Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
							0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
					Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	A	Isolation	III	S											
		Initial			10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
		End-of-life			1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
	B1-B14	V_{OUT} (low-loaded)	IV	S	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	V
	C1,C2, C25,C26	Input breakdown current	V	S	0.0	200	0.0	200	0.0	200	0.0	200	0.0	200	µA
	C3,C22, C27-C30	Input breakdown current	V	S	0.0	100	0.0	100	0.0	100	0.0	100	0.0	100	µA
	C23,C24	Input breakdown current	V	S	0.0	1.6	0.0	1.6	0.0	1.6	0.0	1.6	0.0	1.6	mA
	D1-D14	Output short circuit current	VI	X	-6	-100	-6	-100	-6	-100	-6	-100	-6	-100	mA
	E1-E14	Output leakage current	VII	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
	F1-F24	Input clamp diode voltage	VIII	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
	G1-G8	Functional test	IX	X	← Output must be as specified in table IX →										
	H1-H14	Static output voltage, low level	X	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
	J1-J14	Static output voltage, high level	X	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
	K1-K4	Input current, high level	XI	X	0	80	0	80	0	80	0	80	0	80	µA
	K2,K3, K25,K26	Input current, high level	XI	X	0	40	0	40	0	40	0	40	0	40	µA
	K5-K24, K27-K30	Input current, high level	XI	X	0	20	0	20	0	20	0	20	0	20	µA
*	L1,L4	Input current, low level	XII	X	0.00	-1.72	0.00	-1.72	0.00	-1.72	0.00	-1.72	0.00	-1.72	mA
*	L2,L3	Input current, low level	XII	X	0.00	-0.87	0.00	-0.87	0.00	-0.87	0.00	-0.87	0.00	-0.87	mA
*	L5-L24	Input current, low level	XII	X	0.00	-0.43	0.00	-0.43	0.00	-0.43	0.00	-0.43	0.00	-0.43	mA

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See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
M	Power supply current	XIII	S	0.1	38.6	0.1	38.6	0.1	38.6	0.1	38.6	0.1	38.6	mA
N1-N14	Transition time (t_{TLH}), C_L = 50 pF	XIV	S	5	50	5	50	5	55	5	50	5	60	ns
N15-N28	Transition time (t_{TLH}), C_L = 150 pF	XIV	Q	10	80	10	80	10	90	10	80	10	100	ns
P1-P14	Transition time (t_{THL}),	XIV	S	3	18	3	18	3	20	3	18	3	22	ns
P15-P28	Transition time (t_{THL}), C_L = 150 pF	XIV	Q	6	40	6	40	6	42	6	40	6	44	ns
Q1-Q14	Propagation delay time from clock (t_{PLH}), C_L = 50 pF	XV	S	3	29	3	29	3	31	3	29	3	33	ns
Q15-Q28	Propagation delay time from clock (t_{PLH}), C_L = 150 pF	XV	Q	4	39	4	39	4	41	4	39	4	43	ns
R1-R14	Propagation delay time from clock (t_{PHL}), C_L = 50 pF	XV	S	4	44	4	44	4	46	4	44	4	48	ns
R15-R28	Propagation delay time from clock (t_{PHL}), C_L = 150 pF	XV	Q	5	54	5	54	5	56	5	54	5	58	ns
S1-S8	Propagation delsy time from set to Q (t_{PLH}), C_L = 50 pF	XVI	S	5	55	5	55	5	57	5	55	5	59	ns
S9-S16	Propagation delay time from set to Q (t_{PLH}), C_L = 150 pF	XVI	Q	5	65	5	65	5	67	5	65	5	69	ns
T1-T6	Propagation delay time from set to $\overline{Q}$ (t_{PHL}), C_L = 50 pF	XVI	S	6	65	6	65	6	67	6	65	6	71	ns

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See footnotes at end of table.

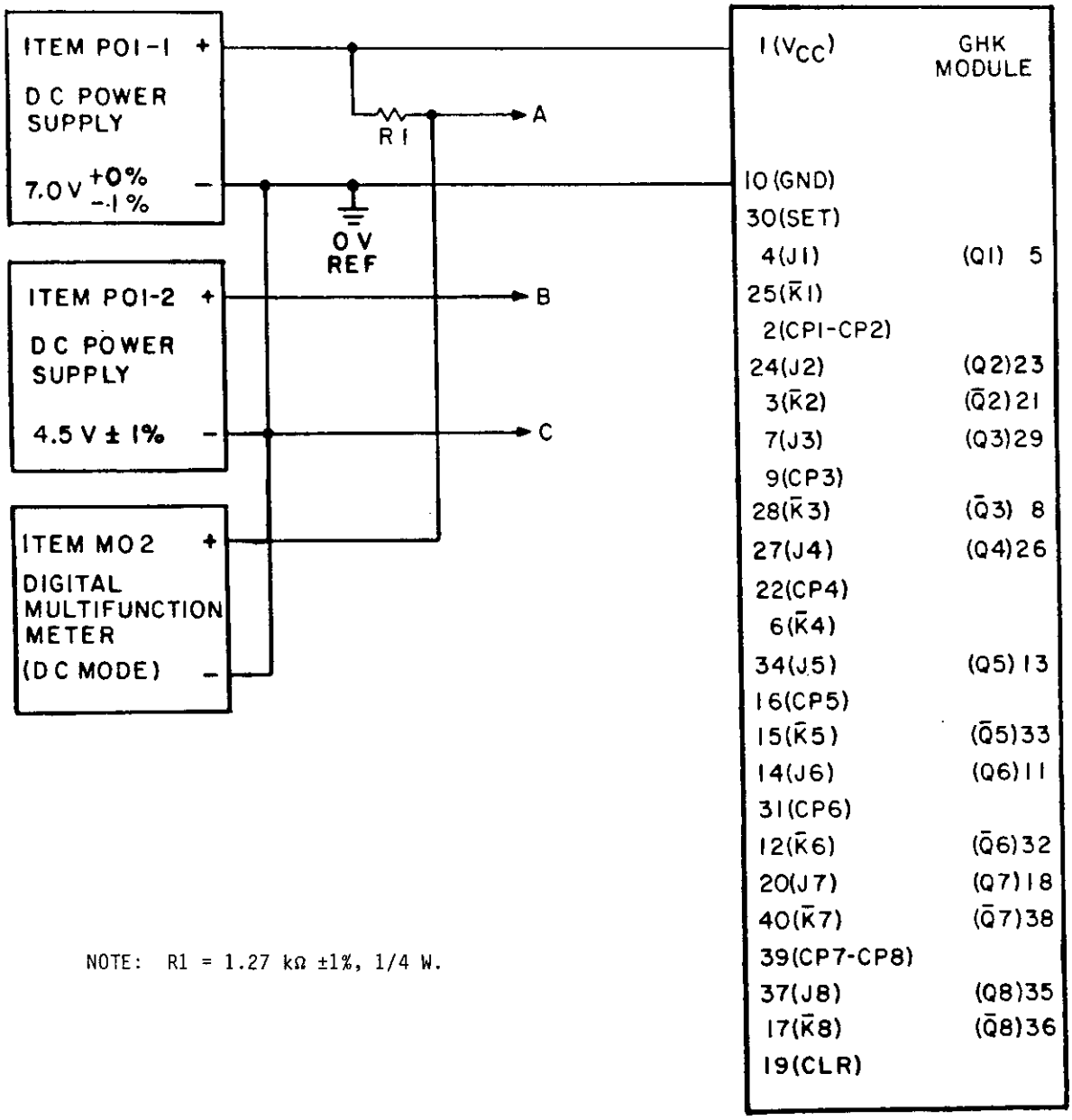
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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
T7-T12	Propagation delay time from set to $\overline{Q}$ (T_{PHL}), C_L = 150 pF	XVI	Q	7	75	7	75	7	77	7	75	7	81	ns
U1-U6	Propagation delay time from clear to $\overline{Q}$ (t_{PLH}), C_L = 50 pF	XVI	S	5	50	5	50	5	52	5	50	5	56	ns
U7-U12	Propagation delay time from clear to $\overline{Q}$ (t_{PLH}), C_L = 150 pF	XVI	Q	6	60	6	60	6	62	6	60	6	66	ns
V1-V8	Propagation delay time from clear to $\overline{Q}$ (t_{PHL}), C_L = 50 pF	XVI	S	6	63	6	63	6	67	6	63	6	71	ns
V9-V16	Propagation delay time from clear to $\overline{Q}$ (t_{PHL}), C_L = 150 pF	XVI	Q	7	73	7	73	7	77	7	73	7	81	ns
W1-W14	Maximum operating frequency check, C_L = 50 pF	XVII	S	110	120	110	120	110	120	110	120	110	120	ns
X1-X14	Output crosstalk, low level	XVIII	Q	0	400	0	400	0	400	0	400	0	400	mV
Y1-Y14	Output crosstalk, high level	XVIII	Q	0	400	0	400	0	400	0	400	0	400	mV
Z	Power supply transient amplitude	XIX	Q	0	400	0	400	0	400	0	400	0	400	mV
AA	Power supply filter capacitance	XX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
BB1	Input capacitance	XXI	Q	2	45	2	45	2	45	2	45	2	45	pF
BB2-BB4	Input capacitance	XXI	Q	2	30	2	30	2	30	2	30	2	30	pF
BB5-BB24	Input capacitance	XXI	Q	2	15	2	15	2	15	2	15	2	15	pF
CC1-CC28	Setup time	XXII	S	170	230	170	230	170	230	170	230	170	230	ns
DD1-DD28	Hold time	XXIII	S	170	230	170	230	170	230	170	230	170	230	ns

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

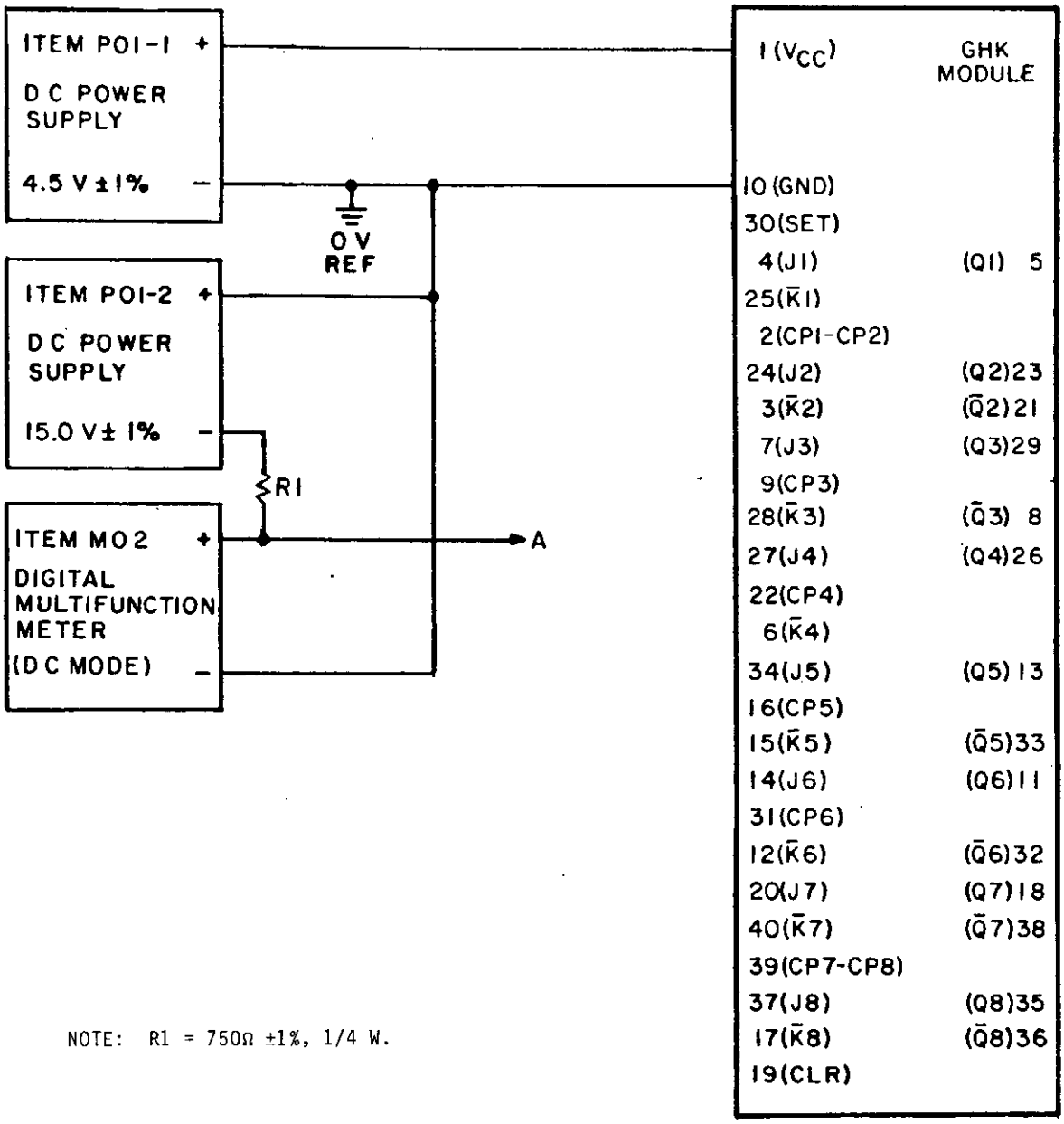
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NOTE: R1 = 1.27 kΩ ±1%, 1/4 W.

FIGURE 7 V_{OUT} (low-loaded) test circuit for table IV.

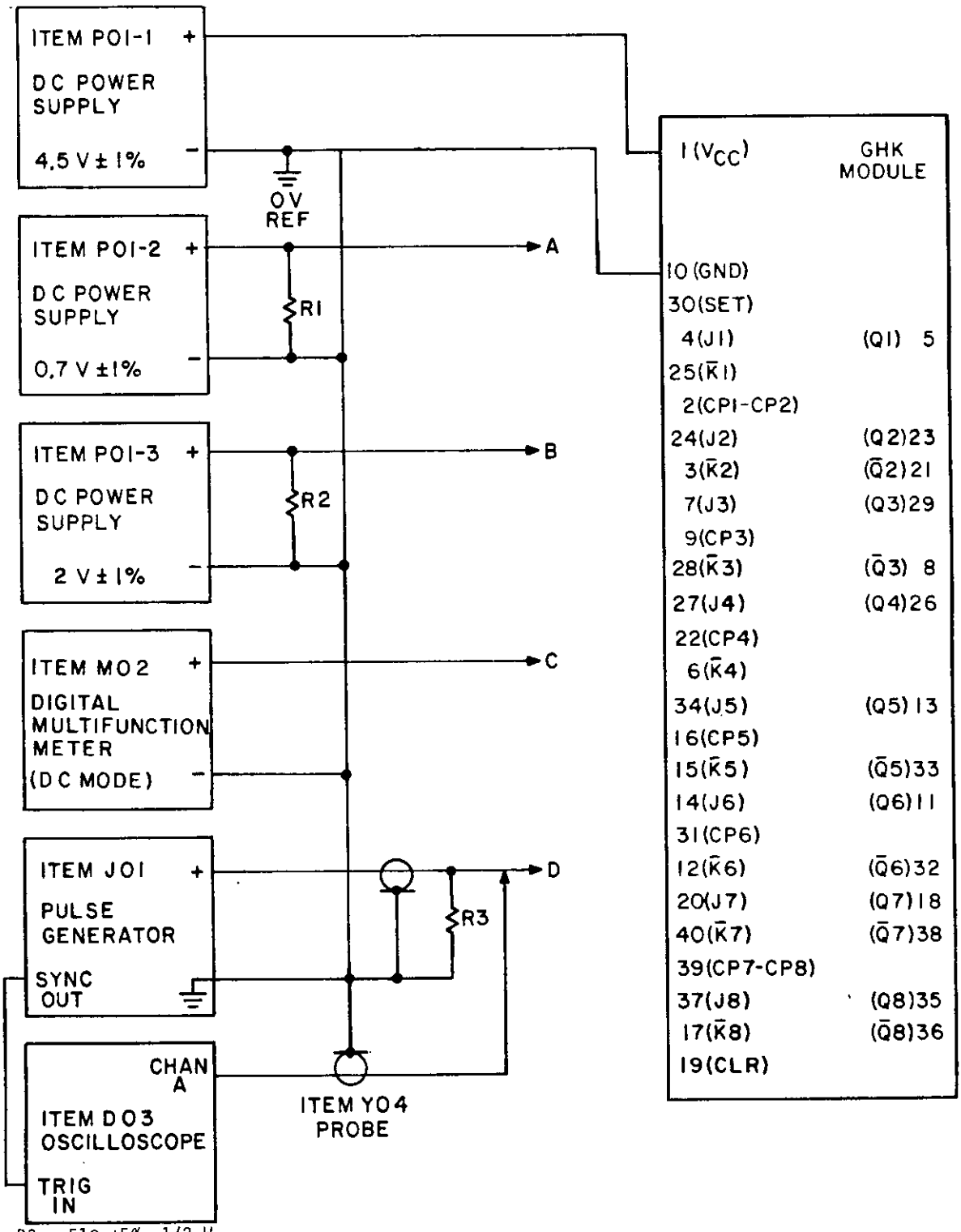
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NOTE: R1 = 750Ω ±1%, 1/4 W.

FIGURE 11. Input clamp diode voltage test circuit for table VIII.

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NOTES:
1. R1 - R3 = 51Ω ±5%, 1/2 W.
2. For test conditions G1 - G8 and J1 - J14, make all input circuit connections for a test condition then use the single pulse mode on item J01 (after initial setup) and apply a single pulse before making each measurement.
3. For test conditions H1 - H14 and J1 - J14, load the output under test and the corresponding complement output (where available) with the load circuit per figure 14. Load circuits are not required for test conditions G1 - G8.

FIGURE 12. Functional and static output voltage (low-level and high-level) and input threshold check test circuit for tables IX and X.

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